

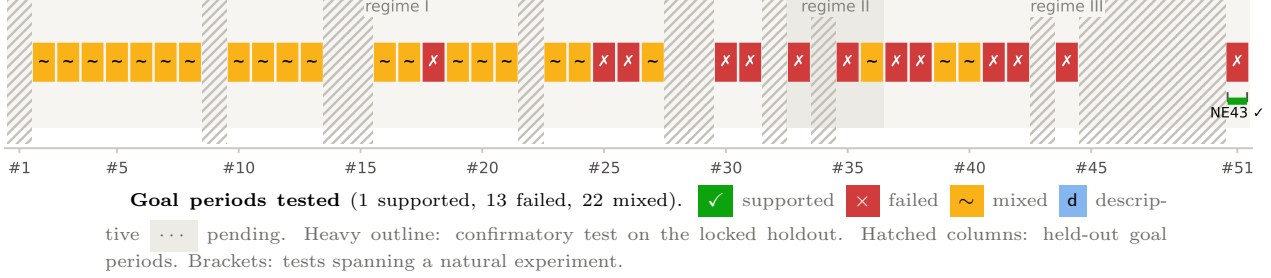
H25: A distance-to-criticality dial, T/T_c , per window and per channel

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. A daily Curie–Weiss dial is computable from logs with honest intervals and, on corrected data, agrees with H19 and H03 and shows activity and talk far from criticality ($g \approx 0.16$, activity 0.07 after trimming), but it is blind to delayed coupling and its near-critical content reading is size times a shared field.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Built a reusable daily dial: the Curie–Weiss loop gain $g = 1 - 1/VR$ and $T/T_c = 1/g$, per day and channel, from logs, with stalls masked, slow drives removed and calibrated 90% intervals; ran it on 282 non-holdout days. **Round 1b** re-ran it on the corrected activity table (round 1 had lost half of all events) with DQ8’s calibrated null. **Activity and talk are subcritical on every day** (median $g = 0.16$ both, was 0.10 and 0.12; $T/T_c \approx 6$); trimmed to the common running window, activity drops to 0.07 and beats the independent-agent ceiling on only 24% of days. The dial now reproduces H19 on the same table ($\rho = 0.94, 0.97$) and tracks H03’s Hawkes $\hat{n}$ ($\rho = 0.46$). **Content still reads near-critical** (0.70; P1 failed), as swarm size times a shared-field correlation. Native tests: withdrawing the operator’s drives (NE43) leaves the dial unchanged; talk co-activation vanishes when 15 agents share one room (NE42). Holdout not run.

PHYSICAL MODEL

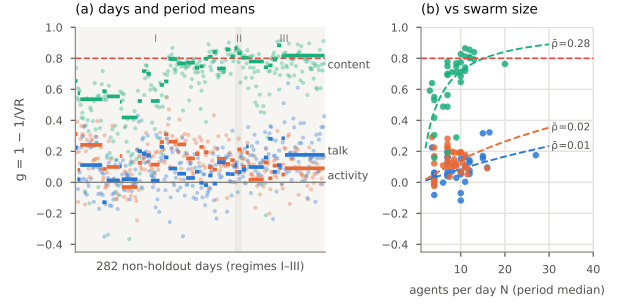
Models 01 (Curie–Weiss), 09 (Hawkes), 11 ($O(n)$ spins). $s_i = \pm 1$: agent i active, or posting, in a minute; for content, s_i is its 30-min mean of whitened unit message embeddings ($n = 32$). $h_i(t)$: private fields (schedule, goal), removed by 30-min block means and agent-day means; stalls masked. $q_i = 1 - m_i^2$; g : loop gain; VR: collective over independent variance; $\bar{\rho}$: mean pairwise correlation; N : agents that day.

$$H = -\frac{J_0}{2N} \left(\sum_i s_i \right)^2 - \sum_i h_i(t) s_i, \quad g = \beta J_0 \bar{q}$$

$$\delta M = \frac{\sum_j q_j \beta \delta h_j}{1 - g}, \quad VR = \frac{\text{Var} \sum_i s_i}{\sum_i \text{Var} s_i} = \frac{1}{1 - g}$$

$$g = 1 - VR^{-1}, \quad T/T_c = 1/g \quad (h = 0)$$

$$VR - 1 = (N - 1) \bar{\rho} \Rightarrow g(N) = \frac{(N - 1) \bar{\rho}}{1 + (N - 1) \bar{\rho}}$$



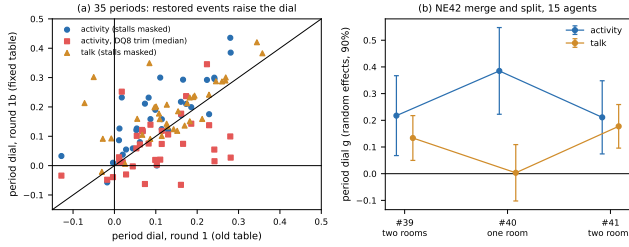
Observables. Round 1 (old table). (a) Daily dial (dots) and period means (bars) for activity (blue), talk (orange) and content (green); red dashes mark $g = 0.8$. Activity and talk stay low; content climbs once rosters grow. On the corrected table activity and talk shift up by about 0.06 (figure on page 2), content is unchanged. (b) Period dial against swarm size: dashed curves hold the per-pair correlation fixed; content and activity follow them (field-like), talk does not (on the fixed table talk $\bar{\rho} \propto N^{-1.25}$).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The useful part is a calibrated instrument and three warnings. Any operator can compute the dial from event logs with `compute_dial`; here it says the swarm amplifies a nudge by 10–20%: no fast runaway. Warning 1: a green dial does not rule out slow herding through context read minutes later. Warning 2: g rises with headcount at fixed per-pair correlation, so track $\bar{\rho}$, weekly. Warning 3 (new): trim each day to the common running window before reading activity feedback; without it, half the “signal” is agents booting together. The talk dial is now the credible gauge: it matches an independent Hawkes fit and survives trimming.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (corrected table)	Verdict
P1 every day subcritical, all channels	activity, talk 100% → 100%; content 22.5% → 24%	×
P2 medians activity 0.03–0.20, talk 0.03–0.25	0.096, 0.122 → 0.159, 0.162	✓
P3a reproduces (same table)	H19 talk ρ 0.69 → 0.97; activity 0.94 → 0.94	×
DQ8 null (trim first)	activity days above ceiling 54% new → 24%; talk 61% → 53%	
P4a talk dial vs H03 ($\rho \geq 0.3$)	$\hat{n}$ 0.25 → 0.46 (fast n_x 0.60)	×
P5 day-to-day signal ($\geq 50\%$)	2/28 → 4/29	×
P7a 03-24: activity up, talk down	+0.08/−0.06 +0.16 [0.02, 0.29]/+0.05	→ ×
NE43 drives withdrawn: no step (native)	$ z \leq 1.47$, inside placebo band	✓
NE42 room merge: size-free (native)	talk g 0.13 → 0.003 → 0.18	~



Second figure. (a) Period dials, round 1 vs 1b: restored events raise activity and talk by about 0.06; trimming (red) pulls activity back down. (b) NE42, 15 agents: talk co-activation vanishes in the one-room week and returns at the split.

GOAL PERIODS IN DETAIL

G02 ~ mixed
G03 ~ mixed
G04 ~ mixed
G05 ~ mixed
G06 ~ mixed
G07 ~ mixed
G08 ~ mixed
G10 ~ mixed
G11 ~ mixed
G12 ~ mixed
G13 ~ mixed
G16 ~ mixed
G17 ~ mixed
G18 × failed
G19 ~ mixed
G20 ~ mixed

+20 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The 1-min dial is a lower bound (context-mediated coupling is invisible); the content dial is an upper bound (shared fields survive). g depends on N . Intervals are widened $\times 1.5$ by synthetic calibration. The fixed-table H19 reference comes from H38's pipeline (same estimator). The #51 size-law test was uninformative; NE42 is goal-confounded.

ROUND 2

Add the DQ8 trim to `confirm.py` Stage B before any hold-out run; build the lagged dial.

- **Essence.** a trustworthy number for how much the swarm amplifies a push; it must be lag-aware and size-aware, with per-pair correlation as the transferable quantity.
- **H25-R1** A lagged (DC) dial from 1–15 min cross-covariances, validated on the delayed-coupling swarms where the 1-min dial reads 0.
- **H25-R2** Agent-level availability masks from H38's scaffold states (consolidation, day edges, infra errors).
- **H25-R3** Content fields from real anchors (goal and kick-off embeddings, artifact events) plus an embedding swap; coordinate with H26.
- **H25-R4** Test the size law $g(N) = (N-1)\rho/(1+(N-1)\rho)$ at roster changes and on the holdout (CA4).
- **H25-R5** A weekly dial with change-point detection, at least 5 agents per day.